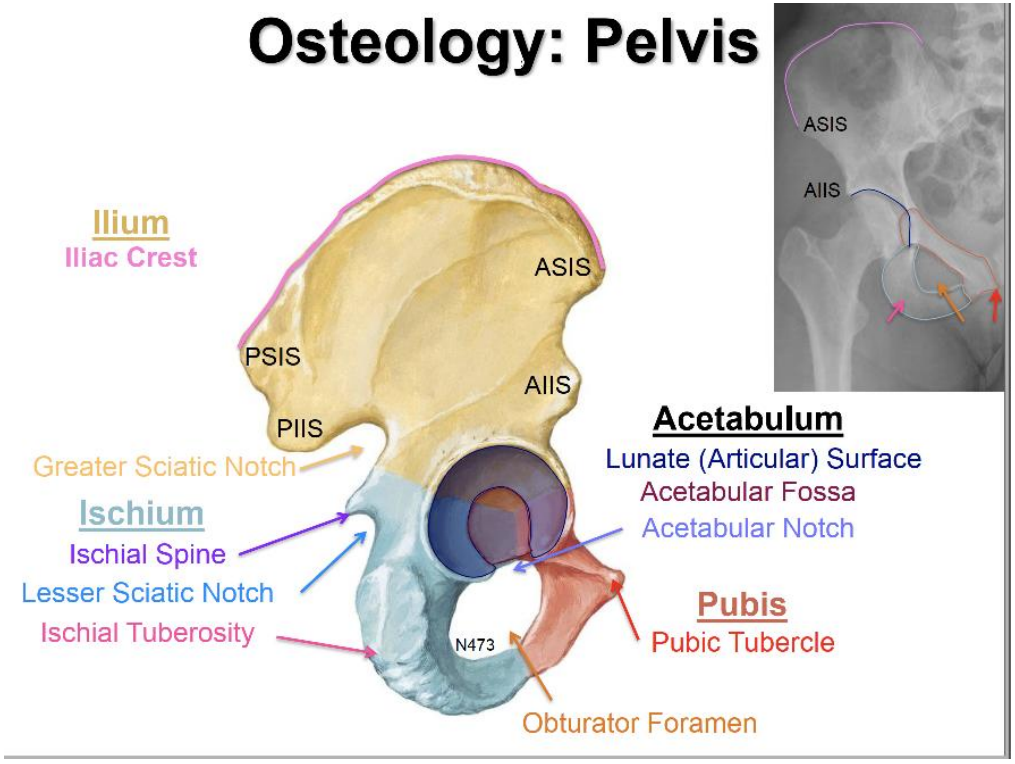
Week #9: [Hip Joint]

Osteology & Joint Features of the Hip

Hip Fractures **(Dr. Cassandra Stiller-Moldovan)**

- Traumatic vs Atraumatic (stress fracture)
- Epidemiology
- Risk Factors
- Clinical Presentation
- Physical Exam
- Investigations
- Fracture types
- Management
- Prognosis

Osteology & Joint Features of the Hip



I. Osteology:

a. **Hip Bone** – comprised of three bones: ilium, ischium, and pubis
   i. **Acetabulum** –articulates with the head of the femur to form the hip joint; all 3 bones (ilium, ischium, and pubis) contribute to the acetabulum

b. **Obturator Foramen** – large oval opening inferior to acetabulum bounded by the ischium and pubis; covered by the obturator membrane; allows passage of the obturator nerves and vessels

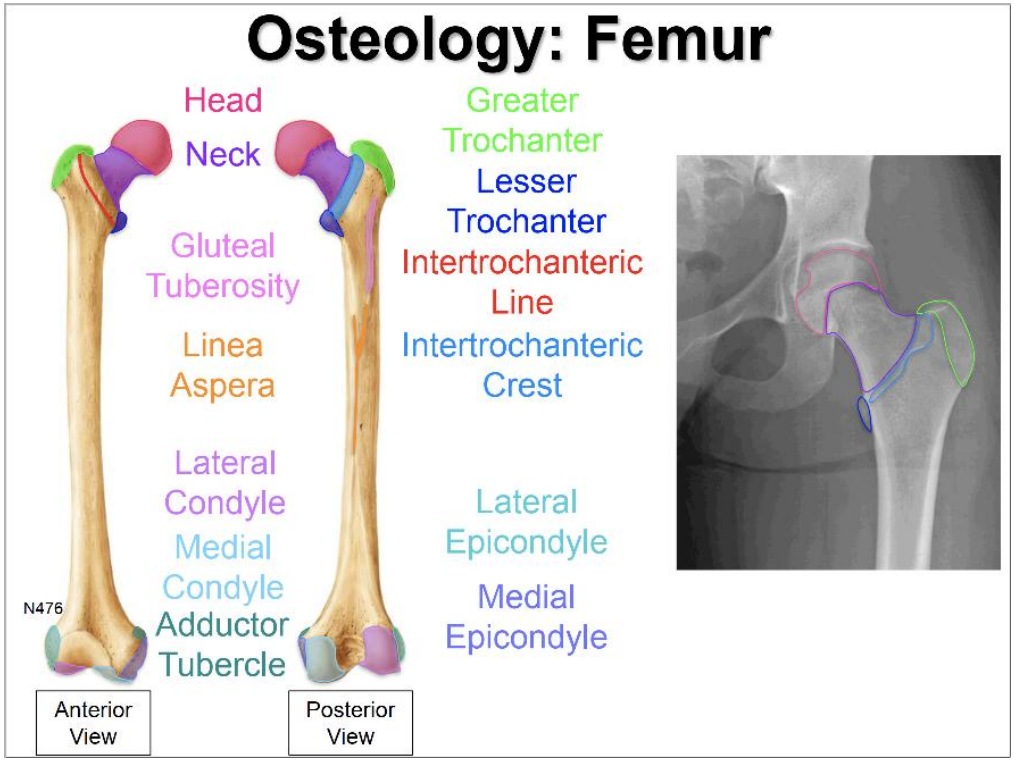
c. **Ilium** – largest part of the hip bone; forms the superior part of the acetabulum

i. **Anterior, Posterior, and Inferior Gluteal Lines** – curved lines on lateral ilium; provide boundaries of attachment for gluteus medius and gluteus minimus
ii. **Iliac Crest** – superior border of ilium; posterior part serves as the origin for gluteus maximus. Top of the crest corresponds to the L4 vertebral level, which can be used as a reference point when performing a lumbar puncture.
    1. **Anterior Superior Iliac Spine (ASIS)** – anterior protrusion of iliac crest
    2. **Anterior Inferior Iliac Spine (AIIS)** – inferior to the ASIS
    3. **Posterior Superior Iliac Spine (PSIS)** – posterior protrusion of iliac crest
    4. **Posterior Inferior Iliac Spine (PIIS)** –inferior to the PSIS; forms the superior border of the greater sciatic notch
    5. **Greater Sciatic Notch** – indentation on ilium that is converted to the greater sciatic foramen by the sacrotuberous and sacrospinous ligaments; enables passage of muscles and neurovasculature from the pelvis to the gluteal region

d. **Ischium** –posteroinferior portion of the hip bone; posterior part of the acetabulum
i. **Ischial Spine** – bony spur on the posterior surface of the ischium; attachment for sacrospinous ligament
ii. **Ischial Tuberosity** –posteroinferior protrusion of ischium; weight is placed on the ischial tuberosity when seated; origin of hamstrings
iii. **Lesser Sciatic Notch** – indentation between the ischial spine & ischial tuberosity that is converted to the lesser sciatic foramen by the sacrotuberous and sacrospinous ligaments; enables neurovascular structures to enter the perineum from the gluteal region

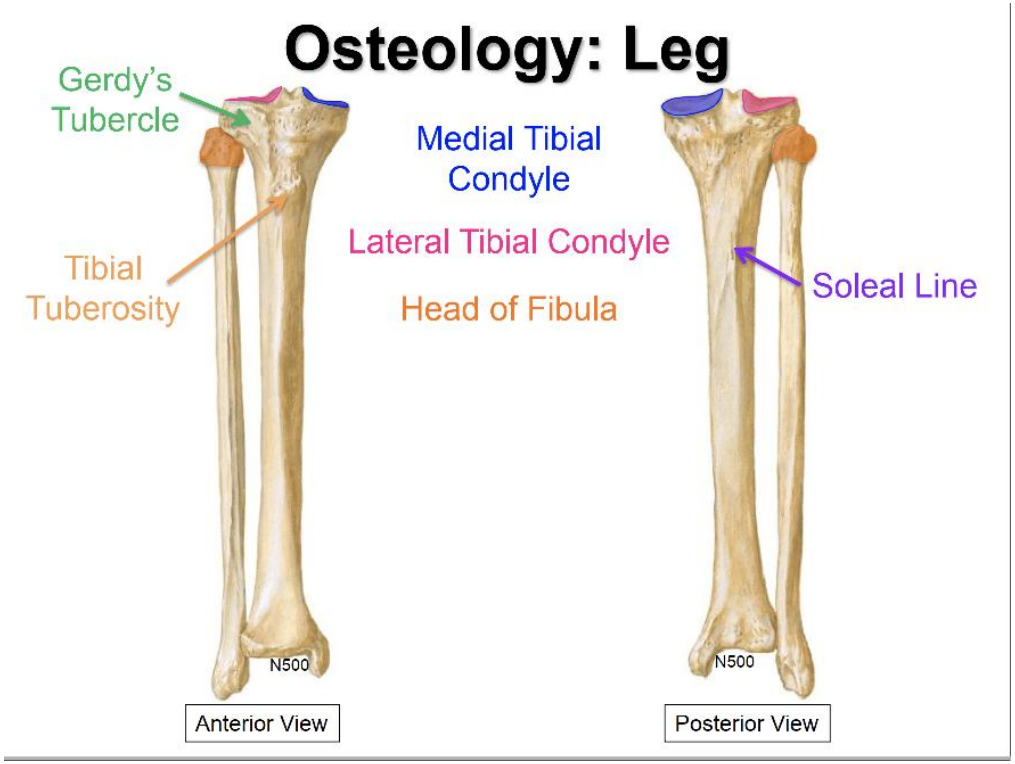
e. **Pubis** –anteromedial portion of hip bone; anterior aspect of the acetabulum
a. **Pubic Tubercle** – anterior bony protuberance; provides an attachment for the inguinal ligament
b. **Superior Ramus** – fuses with the ilium and the ischium at the acetabulum
c. **Inferior Ramus** – fuses with the ramus of ischium to form the inferior aspect of obturator foramen

F. Sacrum – articulates with the ilium to form the sacroiliac joint; serves as an attachment site for the sacrotuberous and sacrospinous ligaments and various muscles of the gluteal compartment



Femur – largest bone in the body located within the thigh; helps transmit weight from the hip to the tibia when standing. The head of the femur articulates with the acetabulum to form the hip joint.

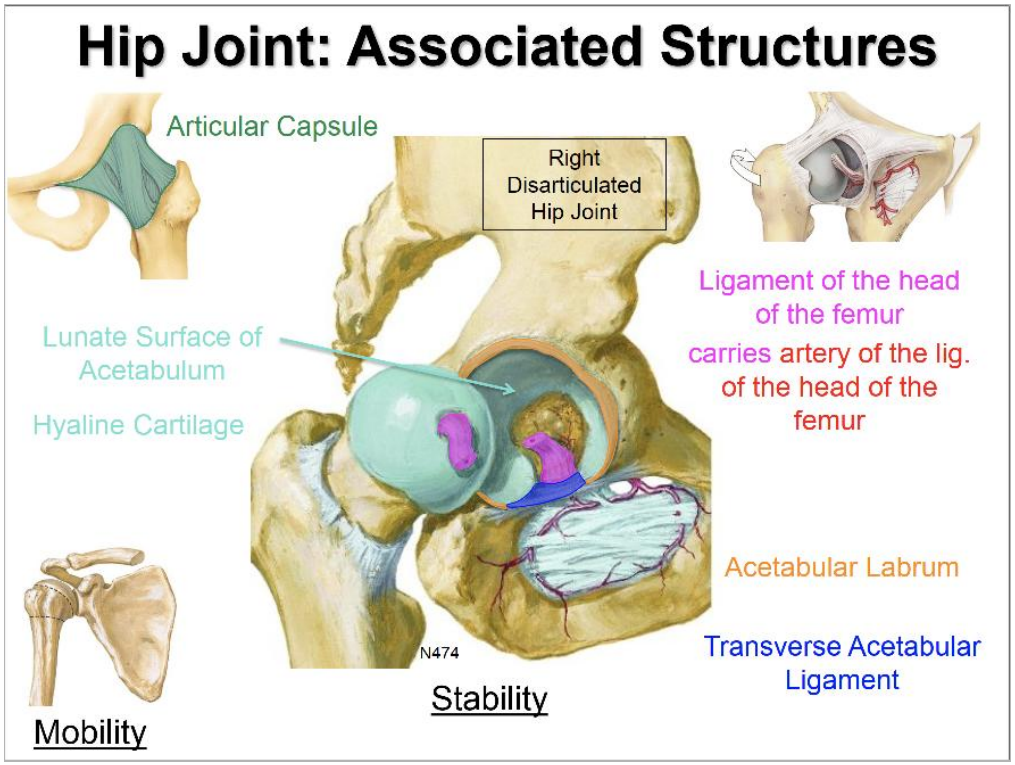
a. **Head** – proximal rounded portion; articulates with acetabulum
b. **Neck** – narrow portion connecting the head and shaft
c. **Greater Trochanter** – large superolateral bony mass at head/neck junction
d. **Lesser Trochanter** – smaller inferior posteromedial bony mass at head/neck junction
e. **Intertrochanteric Line** – anterior line between the greater and lesser trochanters
f. **Intertrochanteric Crest** – posterior ridge between greater and lesser trochanters
g. **Trochanteric fossa** – depression between the neck and greater trochanter
h. **Gluteal Tuberosity** – broad protrusion on posterior aspect of the proximal femur
i. **Linea Aspera** – rough line on the posterior surface of the shaft of the femur
j. **Medial Condyle** – medial distal end of femur; articulates with tibia
k. **Lateral Condyle** – lateral distal end of femur; articulates with tibia
l. **Intercondylar Fossa** – separates the medial and lateral condyles posteriorly



Tibia – major weight bearing bone of the leg

a. **Medial and Lateral Condyle** – articulate with femoral condyles for weight transfer
    i. **Tibial plateau** is formed by the articular surfaces of the two condyles
b. **Tibial Tuberosity** – anterior protrusion; insertion site of the patellar ligament

Fibula – the head of the fibula articulates with the tibia proximally; no weight bearing function

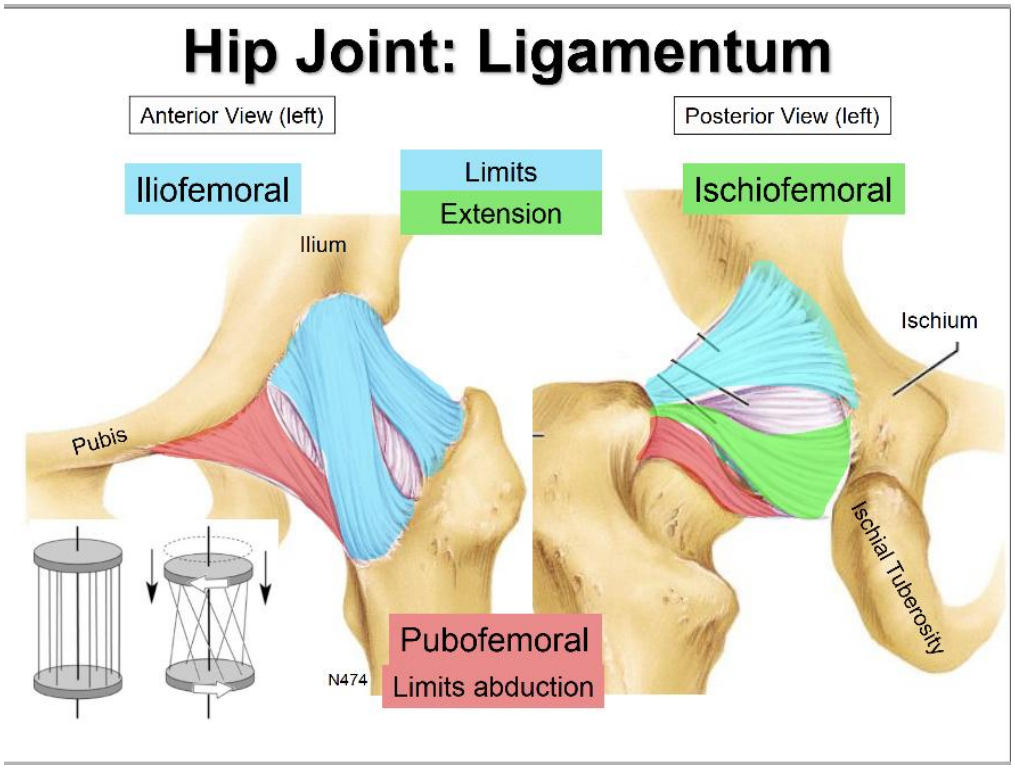


II. Hip Joint:

a. **Most stable joint** in the body; strength conveyed by:
   i. Congruency of fit (greater than 50% of femoral head fits in acetabulum)
   ii. Stout ligaments (including strongest ligament in the body)

Secondary support from extensive surrounding musculature (including dynamic support)

a. ***Type:*** ball-and-socket type of synovial joint; multiaxial
b. ***Articulation:*** acetabulum of hip articulates with head of femur
c. ***Supporting Structures:***
   i. **Joint capsule:** attaches anteriorly to **intertrochanteric line** but does NOT attach posteriorly to femur. Instead, inferior fibers of joint capsule wind posteriorly across the femoral neck (this winding adds dynamic stability to the joint, which actually 'screws' the head of the femur tightly into the acetabulum when the thigh moves into extension and medial rotation).



Thickenings of the capsule give rise to the ligaments of the hip joint:

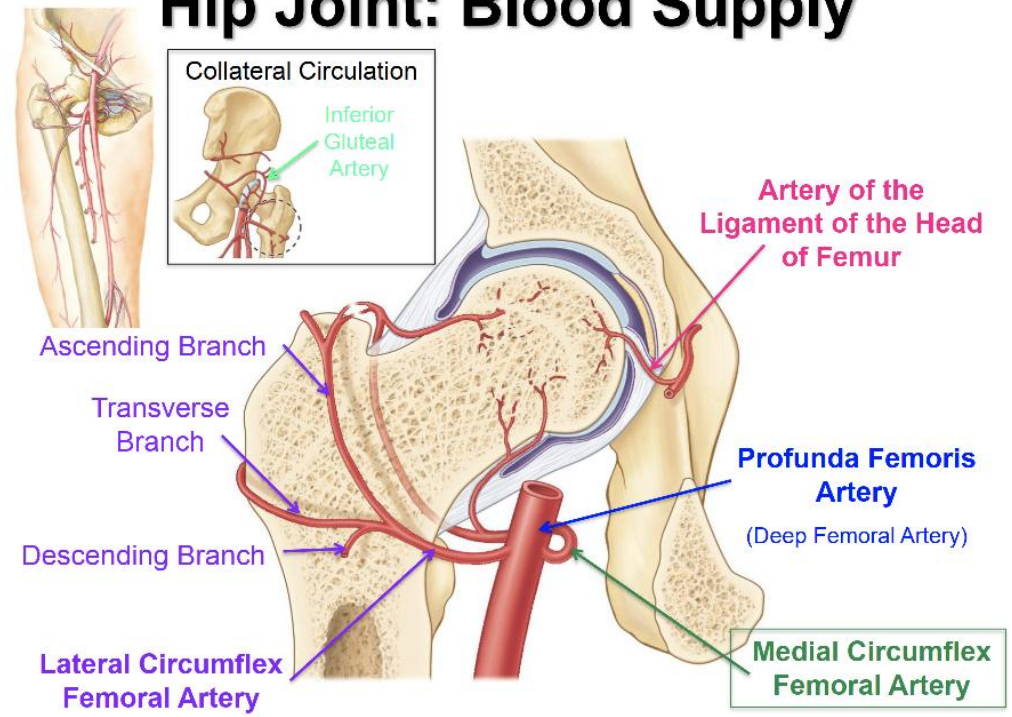
a. **Iliofemoral ligament: "Y ligament of Bigelow"**; strengthens joint anteriorly and superiorly; strongest ligament in the body
b. **Pubofemoral ligament:** strengthens joint anteriorly and inferiorly
c. **Ischiofemoral ligament:** strengthens joint posteriorly

Other structures that increase stability:

a. **Acetabular rim** gives rise to the **acetabular labrum**, a fibrocartilaginous structure that extends from the bony acetabular rim to deepen the 'socket' for the head of the femur.
b. Deficiency in the labrum at the **acetabular notch** is bridged by the **transverse acetabular ligament**

Movements: **multiaxial** and thus **very mobile**; however, the hip joint sacrifices some mobility for stability. Which is in opposition to the glenohumeral joint, which sacrifices stability for mobility.

Hip Joint: Blood Supply



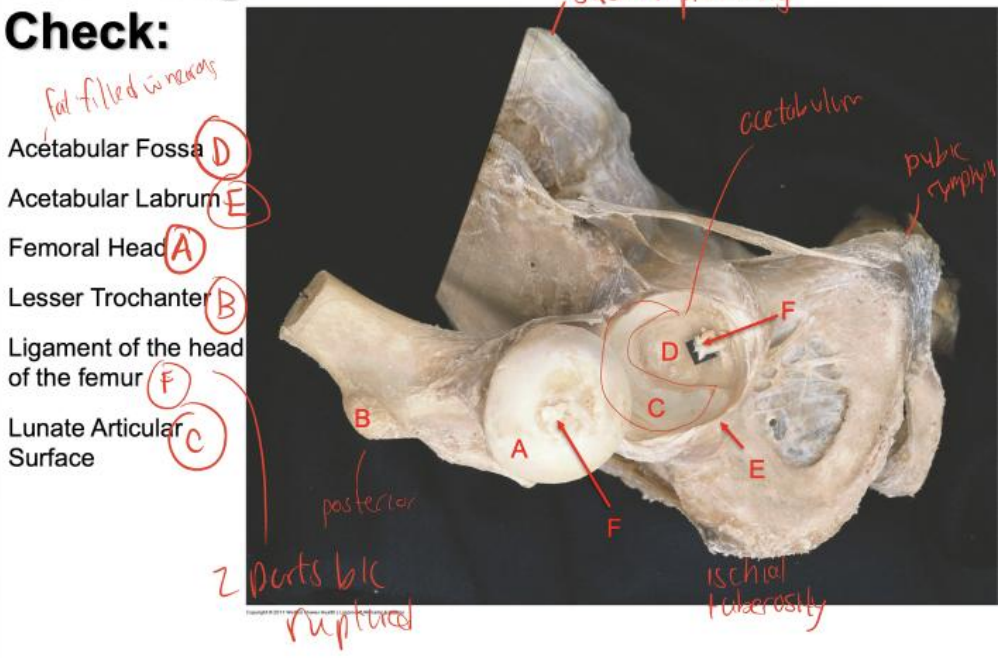
Blood supply:

a. **Medial and lateral circumflex femoral arteries** provide the majority of blood supply (***medial circumflex femoral a***. is the most significant contributor) via retinacular arteries
b. **Artery of the ligament of the head of the femur** travels within the **ligament of the head of the femur**. The artery provides only minor blood flow to the joint.
c. **Note: ligament of the head of the femur** doesn't convey any significant strength to the joint

Clinical Note: The **inferior gluteal artery** anastomoses with branches of the medial and lateral circumflex femoral arteries providing collateral circulation around the hip joint. If there were an occlusion in the proximal femoral artery blood would still be able to supply the lower limb through this anastomosis

Clinical Note: fractures of the neck of the femur can disrupt blood supply from the circumflex arteries resulting in the artery to the ligament of the femoral head being the only remaining blood supply. This artery is frequently inadequate for maintaining the femoral head leading to **avascular necrosis of the femoral head**, likely requiring a total hip replacement. These fractures are especially common in women older than 60 years with osteoporosis

Knowledge Check:



Clinical Note: **Congenital dislocation** of the hip occurs due to the malformation of the acetabulum and/or femur at birth. This bony malformation results in less congruency at the hip joint and reduced stability. **Acquired dislocation** is an uncommon injury to the strength of the hip joint. Posterior dislocation may occur during a car accident resulting in a force being transmitted along the long axis of the femur. Patients with posterior dislocations present with the affected limb medially rotated and appear shorter than the unaffected limb. Because of the close relationship of the sciatic nerve to the hip joint it may become stretched or compressed during dislocation of the hip.

Clinical Note: **Osteoarthritis of the hip** is a common condition affecting the aging population. Osteoarthritis is associated with pathological changes within the joint affecting the articular cartilage, underlying bone, synovium, as well as the surrounding soft tissues. The degenerative pathological changes associated with osteoarthritis can result in a thickened subchondral bone plate, osteophytes, and deformation of the bony morphology. Most conservative treatments for osteoarthritis do not have long-term success; thus, severe osteoarthritis is corrected with a **surgical hip replacement.**

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